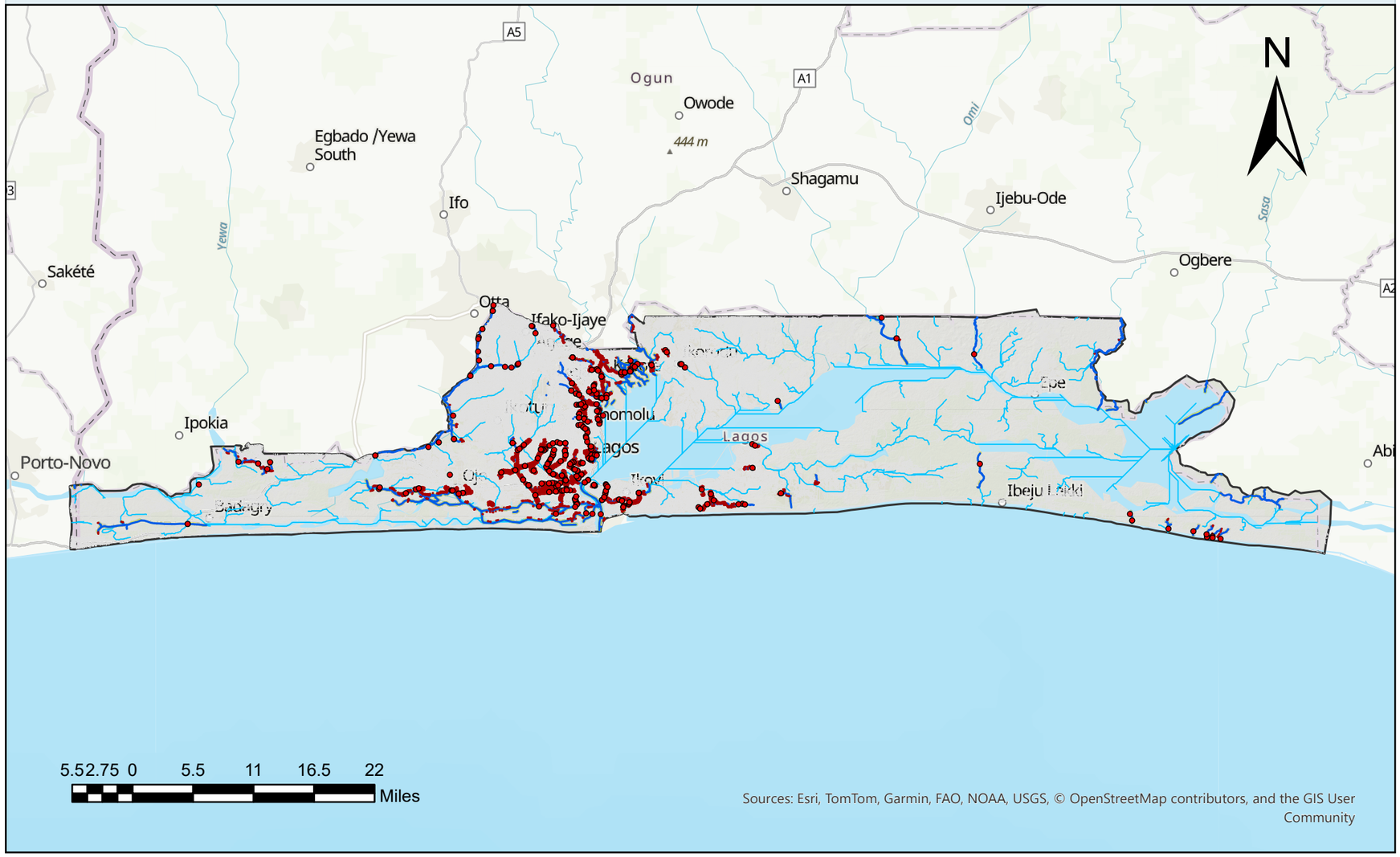
# URBAN DRAINAGE & FLOOD EXPOSURE ANALYSIS

Lagos State, Nigeria | GIS-Based Screening of Low-Lying Terrain, Drainage & Infrastructure Exposure

## MAIN MAP — DRAINAGE CONCERN ZONES & INFRASTRUCTURE EXPOSURE




Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

## KEY RESULTS

**145.92 km²**
Potential drainage-concern area

**3.88%**
Share of Lagos State study area

**780.68 km**
Mapped road length within concern zones

**378**
Road–waterway crossing points

## LEGEND

- Potential Drainage Concern Zone
- Roads Within Concern Zones
- Mapped Waterways
- Terrain-Derived Drainage
- Road–Waterway Crossings
- Lagos State Boundary

## LOCATOR MAP



## METHODOLOGY

SRTM 1 Arc-Second DEM was mosaicked, projected to WGS 1984 UTM Zone 31N and clipped to Lagos State. Elevation, slope, hillshade, flow direction and flow accumulation were derived. OSM roads and waterways were processed to the study area. Low terrain (≤6 m) and a 250 m waterway proximity zone were combined to identify potential drainage-concern areas. Road exposure and road–waterway intersections were then assessed.

## DATA SOURCES

- USGS SRTM 1 Arc-Second Global (~30 m DEM)
- OpenStreetMap / HOT Export — roads & waterways
- geoBoundaries — Lagos State boundary

Software: ArcGIS Pro

## LIMITATIONS & MAP INFORMATION

Screening assessment only — not an official flood-hazard or hydraulic model. Results depend on 30 m SRTM terrain and mapped OSM drainage features. Rainfall intensity, drainage capacity, culverts, underground stormwater systems and flood depth are not modelled.

CRS: WGS 1984 UTM Zone 31N | DEM: 30 m | ArcGIS Pro | Nov 2026